Quantitative Finance

CASE II - Information Entropy and Portfolio Choice

Dan Luo

March 2, 2026

Information Entropy

The concept of *information entropy* was introduced by Shannon (1948)[1] and is also referred to as Shannon entropy.

The entropy of a random variable is the average level of "information", "surprise", or "uncertainty" inherent to the variable's possible outcomes. Given a discrete random variable X, which takes values in $\mathcal{X}$ and is distributed according to $p : \mathcal{X} \to [0, 1]$, the entropy is

$$\mathrm{H}(X) := -\sum_{x \in \mathcal{X}} p(x) \log p(x). \tag{1}$$

The choice of base for $\log$ varies for different applications. Base 2 gives the unit of bits, while base e gives "natural units" (nats), and base 10 gives the unit of "dits".

The *meaning* of the events observed (the meaning of messages) does not matter in the definition of entropy. Entropy only takes into account the probability of observing a specific event.

[1] Shannon, Claude E., 1948. A Mathematical Theory of Communication, *Bell System Technical Journal* 27, 379-423.

Self-Information

The core idea of information theory is that the "informational value" of a communicated message depends on the degree to which the content of the message is surprising.

If a highly likely event occurs, the message carries very little information (e.g., some particular number that will not win a lottery). On the other hand, if a highly unlikely event occurs, the message is much more informative (e.g., some particular number that will win a lottery).

The *information content*, also called the *surprisal* or *self-information*, of an event E is a function which increases as the probability $p(E)$ of an event decreases. When $p(E)$ is close to 1, the surprisal of the event is low, but if $p(E)$ is close to 0, the surprisal of the event is high. This relationship is described by the function

$$\log\left(\frac{1}{p(E)}\right),$$

where $\log$ gives 0 surprise when the probability of the event is 1.

Self-Information - Characterisation

In fact, log is the only function that satisfies a specific set of conditions.

Define an information function I in terms of an event i with probability p_i. The amount of information acquired due to the observation of event i follows from Shannon's solution of the fundamental properties of information:

1. $I(p)$ is monotonically decreasing in p: An increase in the probability of an event decreases the information from an observed event, and vice versa.
2. $I(1) = 0$: Events that always occur do not communicate information.
3. $I(p_1 \cdot p_2) = I(p_1) + I(p_2)$: The information learned from independent events is the sum of the information learned from each event.

Shannon discovered that a suitable choice of I is given by:

$$I(p) = \log\left(\frac{1}{p}\right) = -\log(p). \tag{2}$$

Actually, the only possible values of I are $I(u) = k \log u$ for $k < 0$. Additionally, choosing a value for k is equivalent to choosing a value $x > 1$ for$k = -1/\log x$, so that x corresponds to the base for the logarithm.

Proof.

Let I be the information function which one assumes to be twice continuously differentiable. We have:

$$
\begin{aligned}
\mathrm{I}(p_1 p_2) &= \mathrm{I}(p_1) + \mathrm{I}(p_2) && \text{starting from property 3} \\
p_2\,\mathrm{I}'(p_1 p_2) &= \mathrm{I}'(p_1) && \text{taking the derivative w.r.t } p_1 \\
\mathrm{I}'(p_1 p_2) + p_1 p_2\,\mathrm{I}''(p_1 p_2) &= 0 && \text{taking the derivative w.r.t } p_2 \\
\mathrm{I}'(u) + u\,\mathrm{I}''(u) &= 0 && \text{introducing } u = p_1 p_2 \\
(u\,\mathrm{I}'(u))' &= 0
\end{aligned}
$$

This differential equation leads to the solution $\mathrm{I}(u) = k \log u + c$ for some $k, c \in \mathbb{R}$. Property 2 gives $c = 0$. Property 1 and 2 give that $I(p) \geq 0$ for all $p \in [0, 1]$, so that $k < 0$. $\square$

Entropy

Entropy measures the *expected* (i.e., *average*) amount of information conveyed by identifying the outcome of a random trial. This implies that rolling a die has higher entropy than tossing a coin because each outcome of a die toss has smaller probability ($p = 1/6$) than each outcome of a coin toss ($p = 1/2$).

Consider a coin with probability p of landing on heads and probability $1 - p$ of landing on tails. The maximum surprise is when $p = 1/2$, for which one outcome is not expected over the other. In this case a coin flip has an entropy of one bit. Uniform probability yields maximum uncertainty and therefore maximum entropy. (Similarly, one trit with equiprobable values contains $\log_2 3$ (about 1.58496) bits of information because it can have one of three values.)

The minimum surprise is when $p = 0$ or $p = 1$, when the event outcome is known ahead of time, and the entropy is zero bits. When the entropy is zero bits, this is sometimes referred to as unity, where there is no uncertainty at all - no freedom of choice - no information. Other values of p give entropies between zero and one bits.

Differential Entropy

The Shannon entropy is restricted to random variables taking discrete values. The corresponding formula for a continuous random variable with probability density function $f(x)$ with finite or infinite support $\mathbb{X}$ on the real line is defined by analogy, using the expectation:

$$\mathrm{H}(X) = \mathbb{E}[-\log f(X)] = -\int_{\mathbb{X}} f(x) \log f(x) \, \mathrm{d}x. \tag{3}$$

We replace the summation symbol by an integral. This is the differential entropy (or continuous entropy).

Discussions

The actual continuous version of discrete entropy is the limiting density of discrete points (LDDP). Differential entropy is commonly encountered in the literature, but it is a limiting case of the LDDP, and one that loses its fundamental association with discrete entropy.

The differential entropy lacks many of the properties that make the discrete entropy a useful measure of uncertainty:

- It is not invariant under a change of variables;[2]
- It can become negative (e.g., the uniform distribution $\mathcal{U}(0, 1/2)$ has negative differential entropy since $\int_0^{\frac{1}{2}} -2\log(2)dx = -\log(2)$, while $\mathcal{U}(0, 1)$ has zero differential entropy.);
- It is not even dimensionally correct. Since $h(X)$ would be dimensionless, $p(x)$ must have units of $\frac{1}{dx}$, which means that the argument to the logarithm is not dimensionless as required.

[2] However, it is indeed translation invariance.

Maximum entropy probability distribution

The *principle of maximum entropy* states that the probability distribution which best represents the current state of knowledge about a system is the one with largest entropy, in the context of precisely stated prior data.

A maximum entropy probability distribution has entropy that is at least as great as that of all other members of a specified class of probability distributions. According to the principle of maximum entropy, if nothing is known about a distribution except that it belongs to a certain class (usually defined in terms of specified properties or measures), then the distribution with the largest entropy should be chosen as the least-informative default. The motivation is twofold:

1. Maximizing entropy minimizes the amount of prior information built into the distribution;
2. Many physical systems tend to move towards maximal entropy configurations over time.

Practical Matters

Although there are more general forms of entropy, in connection with maximum entropy distributions, the differential entropy is the only one needed, because maximizing it will also maximize the more general forms.

The natural logarithm and Lebesgue measure are often considered as a “natural” choice.[3]

[3] Information theorists may prefer to use base 2 in order to express the entropy in bits.

For Gross Return

Let $\tilde{R}$ denote the gross return of a stock. Suppose we know that $\tilde{R}$ is supported on $(-\infty, \infty)$. Furthermore, we know its first m moments. That is,

$$\int_{-\infty}^{\infty} R^i f(R) dR = \mu_i, \quad i = 1, 2, ..., m, \tag{4}$$

where $f(R)$ is the continuous density of R.

Of course, we have

$$\int_{-\infty}^{\infty} R^i f(R) dR = \mu_i, \quad i = 0, \tag{5}$$

where $\mu_0 = 1$.

For Gross Return - Cont'd

To find the maximum entropy distribution under the moment conditions, we form the Lagrangian

$$\mathcal{L} = -\int_{-\infty}^{\infty} f(R) \ln f(R) dR + \sum_{i=0}^{m} \lambda_i \left(\int_{-\infty}^{\infty} R^i f(R) dR - \mu_i \right), \tag{6}$$

where the λ_i's are the Lagrangian multipliers.

To apply the calculus of variations (see Part IV), we write

$$\mathcal{L} = \int_{-\infty}^{\infty} \left(-f(R) \ln f(R) + \sum_{i=0}^{m} \lambda_i R^i f(R) \right) dR - \sum_{i=0}^{m} \lambda_i \mu_i, \tag{7}$$

and let

$$F \equiv -f(R) \ln f(R) + \sum_{i=0}^{m} \lambda_i R^i f(R). \tag{8}$$

For Gross Return - Cont'd

We take the functional derivative of F with respect to the unknown $f(R)$ and equate the derivative to 0. We obtain the following F.O.C. (Euler Equation):

$$\frac{\delta F}{\delta f}(f) = -\ln f(R) - 1 + \sum_{i=0}^{m} \lambda_i R^i = 0. \tag{9}$$

$$\Rightarrow \quad f(R) = exp\left(\sum_{i=0}^{m} \lambda_i R^i - 1\right). \tag{10}$$

Suppose we know the first two moments of $\tilde{R}$. Then,

$$f(R) = e^{\lambda_0 - 1} e^{(\lambda_1 R + \lambda_2 R^2)}, \tag{11}$$

which is a normal density. (The λ's are determined by the moment conditions.)

For Gross Return - Cont'd

Now suppose $\tilde{R}$ is supported on $(0, \infty)$. Furthermore, we know:

$$\int_0^\infty (\ln R)^i f(R) dR = \mu_i, \quad i = 1, 2, ..., m, \tag{12}$$

where $f(R)$ is the continuous density of R.

Of course, we have

$$\int_0^\infty (\ln R)^i f(R) dR = \mu_i, \quad i = 0, \tag{13}$$

where $\mu_0 = 1$.

For Gross Return - Cont'd

To find the maximum entropy distribution under the moment conditions, we form the Lagrangian

$$\mathcal{L} = -\int_0^\infty f(R)\ln f(R)dR + \sum_{i=0}^m \lambda_i \left(\int_0^\infty (\ln R)^i f(R)dR - \mu_i \right), \quad (14)$$

where the λ_i's are the multipliers.

We rewrite

$$\mathcal{L} = \int_0^\infty \left(-f(R)\ln f(R) + \sum_{i=0}^m \lambda_i (\ln R)^i f(R) \right) dR - \sum_{i=0}^m \lambda_i \mu_i, \quad (15)$$

and let

$$F \equiv -f(R)\ln f(R) + \sum_{i=0}^m \lambda_i (\ln R)^i f(R). \quad (16)$$

For Gross Return - Cont'd

We take the functional derivative of F with respect to the unknown $f(R)$ and equate the derivative to 0. We obtain the following F.O.C.:

$$\frac{\delta F}{\delta f}(f) = -\ln f(R) - 1 + \sum_{i=0}^{m} \lambda_i (\ln R)^i = 0. \tag{17}$$

$$\Rightarrow \quad f(R) = exp\left(\sum_{i=0}^{m} \lambda_i (\ln R)^i - 1\right). \tag{18}$$

Suppose we know the first two moments of $\ln \tilde{R}$. Then,

$$f(R) = \frac{e^{\lambda_0 - 1}}{R} e^{((\lambda_1+1)\ln R + \lambda_2 (\ln R)^2)}, \tag{19}$$

which is a lognormal density.

Portfolio Choice

In practice, the mean-variance optimization tends to concentrate large-percentage allocations on few assets, often ones with high risk. This creates a sparse solution with little diversification, which is a consequence opposed to the original intention.[4]

Note that under short-sale constraints, the weights of individual assets are positive and sum up to one. Therefore, they can be treated as probability weights. And we further use their entropy as a measure of portfolio diversity.

[4] See Black (1992), Green (1992), Corvalán (2005), and Koumou (2019).

Maximum Diversity

$$\min_{w_1,w_2,\dots,w_N} \quad \sum_{i=1}^{N} w_i \ln w_i, \tag{20}$$

$$s.t. \quad \sum_{i=1}^{N} w_i \bar{r}_i = \bar{r}_p, \tag{21}$$

$$\sum_{i=1}^{N} w_i = 1, \tag{22}$$

$$-w_i \leq 0, \;\; i = 1, 2, \dots, N. \tag{23}$$

Solution

Let

$$\mathcal{L} = \sum_{i=1}^{N} w_i \ln w_i + \lambda \left(\sum_{i=1}^{N} w_i \bar{r}_i - \bar{r}_p \right) + \eta \left(\sum_{i=1}^{N} w_i - 1 \right) - \sum_{i=0}^{m} \zeta_i w_i, \tag{24}$$

where λ, η, and ζ's are the Lagrangian multipliers.

The F.O.C.s are

$$\ln w_i + 1 + \lambda \bar{r}_i + \eta - \zeta_i = 0, \;\; i = 1, 2, ..., N, \tag{25}$$

$$\sum_{i=1}^{N} w_i \bar{r}_i = \bar{r}_p, \tag{26}$$

$$\sum_{i=1}^{N} w_i = 1, \tag{27}$$

$$\zeta_i \geq 0, w_i \geq 0, \zeta_i w_i = 0, \;\; i = 1, 2, ..., N. \tag{28}$$

Solution - Cont'd

From (25), we get

$$w_i = exp(-1 - \lambda \bar{r}_i - \eta + \zeta_i) > 0, \ i = 1, 2, ..., N. \tag{29}$$

Then $\zeta_i = 0$, $i = 1, 2, ..., N$. We can further derive that

$$w_i = \frac{e^{-\lambda \bar{r}_i}}{\sum_{j=1}^{N} e^{-\lambda \bar{r}_j}}, \ i = 1, 2, ..., N, \tag{30}$$

where λ is determined by the equation:

$$\sum_{i=1}^{N} (\bar{r}_i - \bar{r}_p) e^{-\lambda \bar{r}_i} = 0. \tag{31}$$

For more discussions, see: Bera, Anil K., and Sung Y. Park, 2008. Optimal portfolio diversification using the maximum entropy principle, *Econometric Reviews* 27, 484-512.